

PERFORMANCE WORK STATEMENT (PWS)
for
Wisconsin Model V-465D1 Engine Overhauls
Naval Base Coronado (NAS North Island)

A. General:

1. Introduction: The Contractor shall provide all personnel, equipment, tools, materials, supervision, quality control and other items and non-personal services necessary to perform repair service for 65 horse power, gasoline fired internal combustion engines, Electronic ignition, Wisconsin Model V-465D1 as defined in this PWS. The Contractor shall perform to the standards in this contract and return engines to original operating specifications. Up to 5 engines repaired in the base year and 5 engines repaired in each subsequent option year.

2. Background: Commander Navy Region Southwest (CNRSW), has Wisconsin (Model V-465D1) engines in need of overhaul repair. The engines are diagnosed with a range of problems including: seized pistons, scuffing of the cylinders and debris from the pistons in the bearings. A full list of required parts and labor is detailed in section E.1.

Engines will be delivered to contractor facility as they fail and are identified in need of overhaul. Overhaul and repair of each engine to be completed within 30 days after delivery to contractor facility.

B. General Information:

1. Place and Performance of Services: Contractor's repair facility.

2. Identification of Contractor Employees: All contractor personnel shall wear identifiable clothing and display their temporary badge (if visit to Government property becomes necessary).

3. Designated Representative: The requiring activity representative is Mr. Mark Sojourner, Commander Navy Region Southwest Air Operations Director. Telephone: 619-705-5305; email: mark.e.sojourner.civ@us.navy.mil

C. Government Furnished Property, Material, Equipment and Services (GFP/M/E/S):

The engines are located at various locations and shall be provided to contractor after contract award. The engines will be delivered in shipments of 1 each, during the period of performance, with at least 10 days notice to contractor. Last engine will be delivered NLT 45 days prior to end of period of performance

D. Contractor Furnished Property, Materials, Equipment (CFP/M/E):

All tools and equipment necessary for repair of engine.

E. Contractor Requirements

1. The Contractor shall perform the following repairs and replacement:

New Parts

- Pistons & pins
- Piston rings
- Rod, main, & cam bearings
- Timing kit – domestic OHV
- Rod bearings (if used)
- Brass freeze plug kit
- Cam & oil galley plugs
- Camshaft-reground-new available
- New valve springs (as required)
- New valve lifters
- New exhaust valves
- Rockers & push rods (as required)
- Unleaded exhaust seats (as required)
- Oil pump
- Full gasket set
- Timing kit - OHC

Labor

- Clean & magnaflux block
- Bore cylinders
- Power hone cylinders
- Surface blocks
- Align hone main saddles
- Resize connecting rods and align
- Rebush rods (as required)
- Grind crankshaft
- Install all oil & freeze plugs
- Install cam bearings
- Assemble short block
- Clean & magnaflux heads
- Three angle valve job
- Repair all guides (as required)
- Surface heads
- Install heads-lifters-valves
- Train & adjust
- Paint exterior to resemble “like new” condition

2. Repair work to be completed in accordance with manufacturer’s technical manuals and drawings provided by Government. Repair work shall include a minimum one-year warranty on parts and labor, which commences upon acceptance by government representative.

3. Inspection and acceptance shall be performed at the Contractor’s repair facility by a government representative. Upon satisfactory completion of acceptance inspection, contractor will crate the engines appropriately for medium to long-term storage and government representative will transport engines back to the appropriate installation.

4. Deliverables:

- 4.1 Minimum Performance Specifications (see NAVAIR Overhaul guide, Reference 1). No test run is expected, however the engine crankshaft shall turn and make compression.
- 4.2 Engines shall be returned to the government dry, with oil drained (only assembly lubrication remaining from the rebuilding process). A new filter does not need to be installed for shipping. Cover all openings covered to preserve integrity and crate for storage (unless otherwise instructed).
- 4.3 The fuel strainer shall be cleaned and replaced
- 4.4 Bore cylinder and grind crankshaft as necessary to return engine to operating condition.
- 4.5 Reuse all fasteners, oil lines and fuel lines unless unusable. Replace unusable items.
- 4.6 Oil pressure relief valve shall be replaced.

5. Additional Details:

- Engines are delivered with various external accessories removed. External accessories and lines are not expected to be replaced; only internal rework is required.
- The following items are not part of this servicing requirement:
 - Fuel pump
 - Governor
 - Heat Deflectors
 - Starter
 - Starter Switch
 - Starter Gear
 - Accessory Drives
 - Alternators
 - Gauge
 - Switch Panel
 - Oil Pressure
 - Ammeter
 - Battery Voltage
 - Ignition Switch
 - Ignition Systems
 - Distributor
 - Points (part of distributor)
 - Electronics
 - Plug wires
 - Plugs
 - Ignition coil

F. Reports- Provide summary of repairs completed upon delivery of engine to US Government.

G. Method of payment-Method of payment shall be via Wide Area Workflow.

H. Instructions/References:

- (1) NAVAIR 51-5-31 Section X, Overhaul Instructions
- (2) Wisconsin Models V465D, V460D, V461D Repair Manual
- (3) Wisconsin Motors Canada Quick Reference Specifications
- (4) Wisconsin V456 Manufacturers Parts Breakdown
- (5) Engine Photo 1
- (6) Engine Photo 2
- (7) Engine Photo 3